

Disease X

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Introduction

Disease X is a respiratory virus disease that is rapidly spreading. The outbreak has begun in Berlin, and is beginning in Ho Chi Minh City. The WHO wants to know if disease X will be contained and what can be done to halt the disease becoming the epidemic and minimize the severity of the disease.

This report proposes the simplified mathematical model of Disease X to investigate its infection dynamics and explore the potential control measures that can halt the epidemic and minimize the severity of the disease.

Background

Note

Transmission dynamics:

- Urban areas of Berlin saw higher transmission rates (0.75 per day).
- Rural areas had 30% of the urban transmission rate.
- In Ho Chi Minh City, ~60% of the population is lower risk with similar transmission differences.

Current infection status in Ho Chi Minh City:

- 5 suspected cases, 2 confirmed cases in a 3-million population.
- No latent exposed individuals assumed.

Disease progression & infectiousness:

- 10% of exposed individuals become infectious before symptoms.
- Latency period: ~3 days before symptoms appear.
- Severe cases remain infectious for ~8 days before recovery with assumed long-term immunity.
- Diagnosed individuals recover in ~5 days with immunity.

Diagnosis & treatment:

- Antivirals reduce infectiousness to 38% of undiagnosed individuals.
- Diagnosis rate in Berlin: 1/4 per day; Ho Chi Minh City estimated to be 1/3 per day.

Mortality:

- Estimated at 0.006 per day, same for both diagnosed and undiagnosed individuals.

Model 1 - Basic model with quarantine

This is the basic compartmental model scheme of Disease X, constructed with the given information.

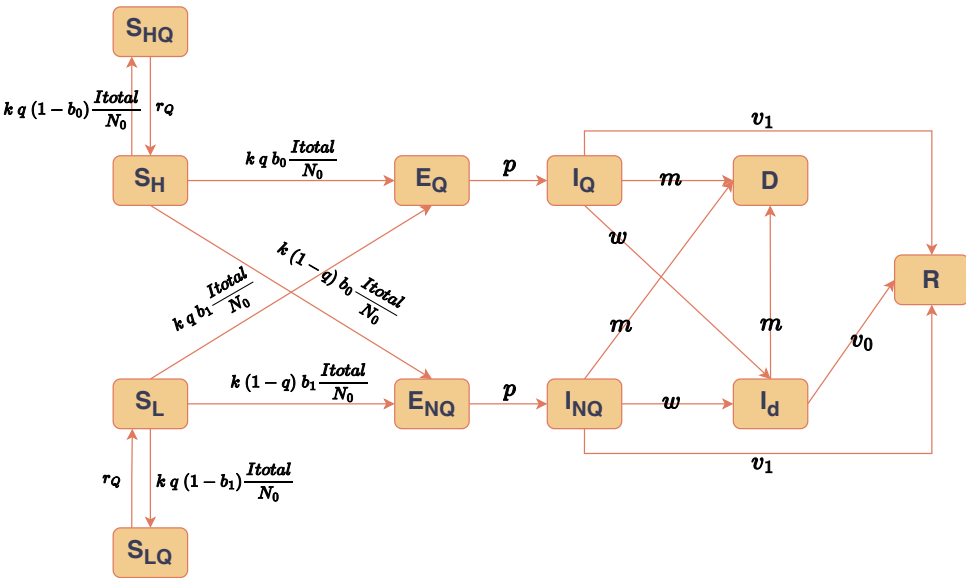


Fig 1. Model Scheme of Disease X without quarantine

Symbol	Definition
S_{HQ}	Susceptible in high-risk population, quarantined
S_H	Susceptible in high-risk population (not quarantined)

Symbol	Definition
S_{LQ}	Susceptible in low-risk population, quarantined
S_L	Susceptible in low-risk population (not quarantined)
E_Q	Exposed, quarantined
E_{NQ}	Exposed, not quarantined
I_Q	Infected, quarantined
I_U	Infected, not quarantined
I_D	Infected, diagnosed (quarantined at this point)
D	Death
R	Recovered

Parameter	Definition	Value
α	Average time until immunity is lost	180 days (rate = $\frac{1}{180}$ day ⁻¹)
k	Baseline daily number of contacts per capita	10 (day ⁻¹)
b_0	Transmission probability per contact in high-risk population	0.075
b_1	Transmission probability per contact in low-risk population	$0.075 \times \frac{1}{3}$
p	Progression rate from exposed to infectious	$\frac{1}{3}$ (day ⁻¹)
v_0	Per capita recovery rate with treatment	$\frac{1}{5}$ (day ⁻¹)
v_1	Per capita recovery rate without treatment	$\frac{1}{8}$ (day ⁻¹)
m	Per capita death rate	0.006 (6 out of 1000 dies daily)
w	Daily rate of diagnosis and access to proper treatment	$\frac{1}{3}$ (day ⁻¹)
r_Q	Average quarantine duration for susceptible, unexposed and uninfected individuals	14 days (rate = $\frac{1}{14}$ day ⁻¹)
q	Fraction of uninfected & unexposed individuals being quarantined	0 to 1

We consider two distinct populations: a high-risk population (S_H) and a low-risk population (S_L), which, for simplicity, are referred to as urban and rural populations, respectively. A portion of individuals in both S_H and S_L who have not yet been exposed to the virus are quarantined, represented mathematically as $kq(1 - b)I_{\text{total}}/N_0$.

Those who are not quarantined are susceptible to virus exposure. Upon exposure, individuals in S_H and S_L transition to either a quarantined exposed state (E_Q) or a non-quarantined exposed state (E_{NQ}). Regardless of quarantine status, once exposed, it takes a period p for individuals to become symptomatic and fully infectious. At that point, E_Q and E_{NQ} progress to infectious states—either quarantined (I_Q) or non-quarantined (I_{NQ}). From there, individuals face three possible outcomes: death (at a rate m), recovery without treatment (at a rate v_1), or proper diagnosis and treatment (at a rate w). Once properly treated, individuals recover at a faster rate v_0 .

After diagnosis and proper treatment (denoted as I_d), individuals either recover more quickly (at rate v_0) or still face a risk of death (at rate m).

This conceptual framework is represented mathematically by the equations provided below.

Total infectious: $I_{\text{total}} = 0.10 \cdot E_{NQ} + I_U + 0.38 \cdot I_D$

1.

$$\frac{dS_{HQ}}{dt} = k(1 - b_0)q \cdot \frac{S_H I_{\text{total}}}{N_0} - r_Q S_{HQ}$$

2.

$$\frac{dS_H}{dt} = r_Q S_{HQ} - kq(1 - b_0) \cdot \frac{S_H I_{\text{total}}}{N_0} - k(1 - q)b_0 \cdot \frac{S_H I_{\text{total}}}{N_0} - kqb_0 \cdot \frac{S_H I_{\text{total}}}{N_0}$$

3.

$$\frac{dS_{LQ}}{dt} = k(1 - b_1)q \cdot \frac{S_L I_{\text{total}}}{N_0} - r_Q S_{LQ}$$

4.

$$\frac{dS_L}{dt} = r_Q S_{LQ} - kq(1 - b_1) \cdot \frac{S_L I_{\text{total}}}{N_0} - k(1 - q)b_1 \cdot \frac{S_L I_{\text{total}}}{N_0} - kqb_1 \cdot \frac{S_L I_{\text{total}}}{N_0}$$

5.

$$\frac{dE_Q}{dt} = kqb_0 \cdot \frac{S_H I_{\text{total}}}{N_0} + kqb_1 \cdot \frac{S_L I_{\text{total}}}{N_0} - pE_Q$$

6.

$$\frac{dE_{NQ}}{dt} = k(1 - q)b_0 \cdot \frac{S_H I_{\text{total}}}{N_0} + k(1 - q)b_1 \cdot \frac{S_L I_{\text{total}}}{N_0} - pE_{NQ}$$

7.

$$\frac{dI_Q}{dt} = pE_Q - I_Q(m + w_1 + v_0)$$

8.

$$\frac{dI_U}{dt} = pE_{NQ} - I_U(m + w_2 + v_0)$$

9.

$$\frac{dI_D}{dt} = w_1 I_Q + w_2 I_U - mI_D - v_1 I_D$$

10.

$$\frac{dD}{dt} = m(I_Q + I_U + I_D)$$

11.

$$\frac{dR}{dt} = v_0(I_Q + I_U) + v_1 I_D$$

The assumptions made to construct this model are as follows:

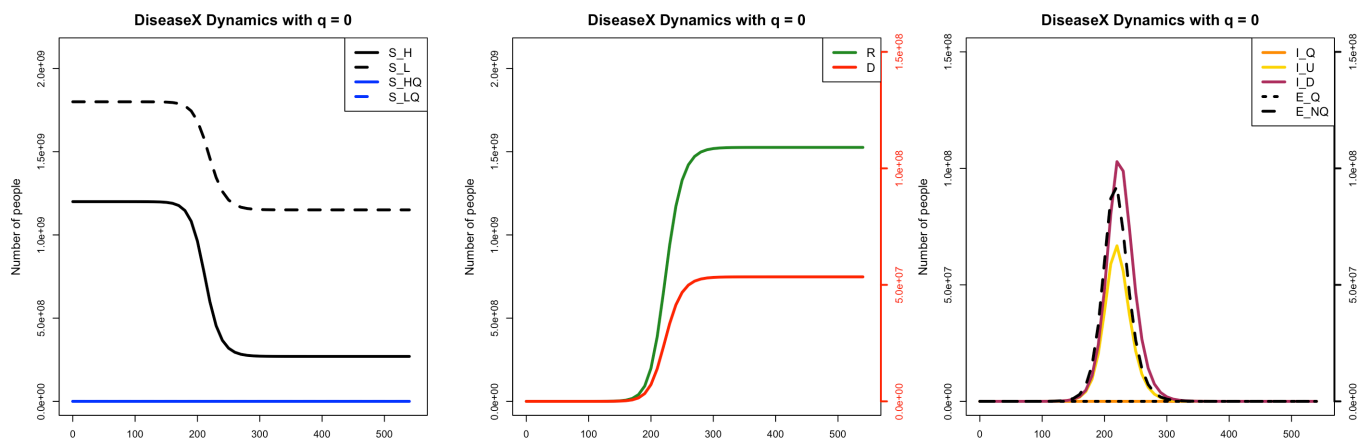
- Metapopulation effects are considered negligible.
- Interactions between the urban and rural populations are assumed to be negligible.
- Seasonal variation in transmission is not included.
- Only individuals who are susceptible and not yet exposed are subject to quarantine.
- Once individuals are exposed, they may either be quarantined (E_Q) or not quarantined (E_{NQ}). Regardless of quarantine status, the diagnosis rate remains the same and is denoted by w .
- If diagnosed, individuals are assumed to receive proper treatment, which leads to faster recovery at rate v_0 , compared to untreated recovery at rate v_1 .
- The model operates under a batch-processing structure, meaning there is no inflow of new individuals (e.g., through births). Consequently, most individuals will eventually transition to either the deceased (D) or recovered (R) compartments, with minimal long-term retention in intermediate states such as exposed (E) or

infectious (I).

Research Question 1. What is the expected dynamics of diseases X?

Without any quarantine or other preventative measure, the disease shows the following dynamic for 1.5 years like below.

- Quarantine level: 0
- Population: 3000000000 (3million)
- Duration: 1.5 year



In the absence of any intervention (i.e., with zero quarantine), the model indicates that the epidemic begins around Day 200 and reaches equilibrium by approximately Day 280. This suggests that, without external control measures, the disease naturally subsides within about 1.5 years.

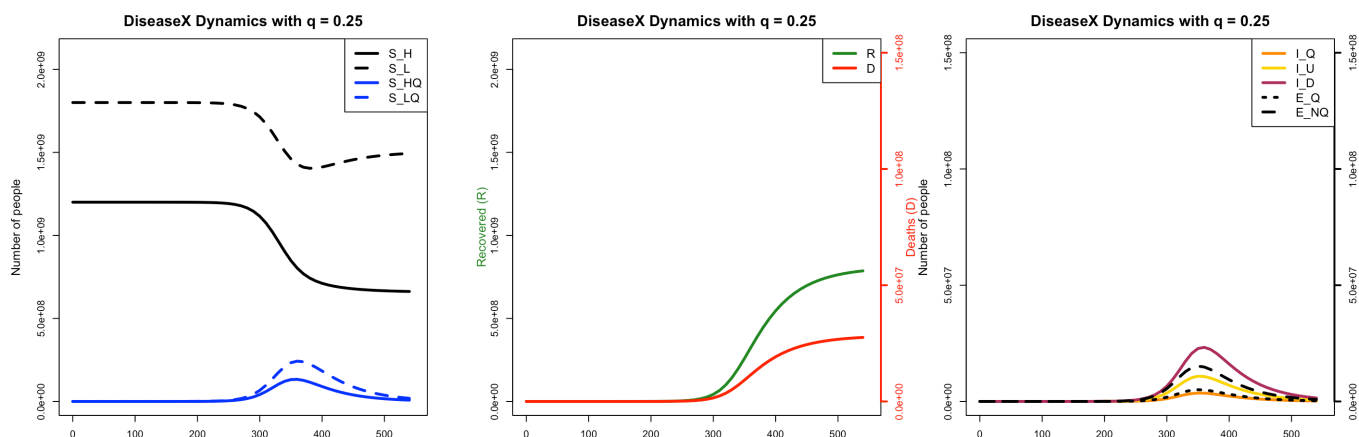
In the 90-day projection, the epidemic appears inactive, as the graph lines remain nearly flat. However, changes are indeed occurring—though minimal and not visually prominent due to the large total population size (3 million). For example, when infections increase from 34 to 45 cases, the change is numerically small and difficult to detect on a population-scale graph. This slow initial progression is expected, given that the outbreak starts with only 5 diagnosed cases in a large population.

This pattern reflects real-world epidemics, such as the early stages of the COVID-19 pandemic. While the outbreak began in China and spread rapidly to neighboring countries like South Korea and Japan, it took several months—approximately six months to a year—for significant transmission to occur in regions like Europe and the United States. A detailed timeline of the COVID-19 pandemic is available here: [Timeline of the COVID-19 pandemic](#).

Research Question 2. What is the effectiveness of quarantine?

Let's introduce the quarantine and see if it has an impact.

- Quarantine level: 25%
- Population: 3000000000 (3million)
- Duration: 1.5 year

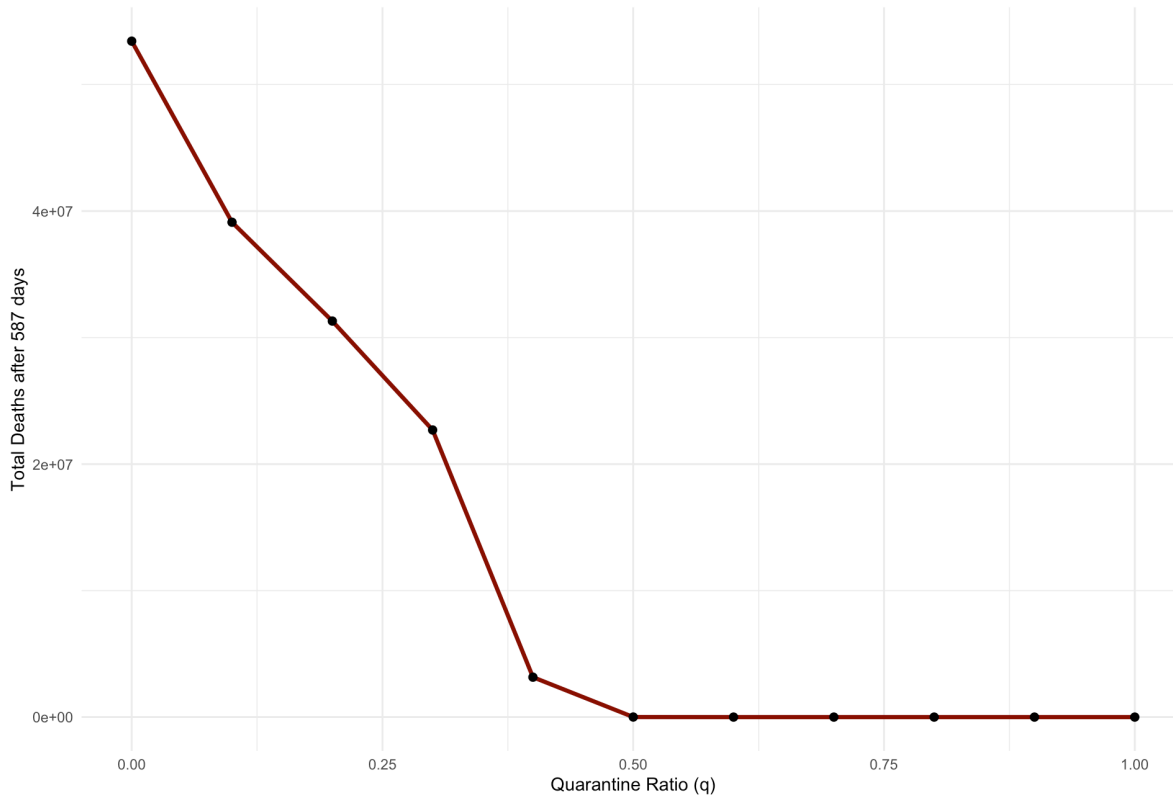


When quarantine measures are introduced, the number of infected individuals decreases significantly. Additionally, the number of people who remain uninfected by the end of the epidemic is higher compared to the scenario with no quarantine. This indicates that quarantine helps prevent exposure, allowing a greater proportion of the population to remain susceptible but uninfected. Consequently, both the number of deaths and recoveries decline, as fewer people contract the disease overall.

This raises an important question: what level of quarantine is most effective? To answer this, a sensitivity analysis is recommended to evaluate how varying degrees of quarantine influence outcomes such as total deaths. This approach can help identify an optimal balance between public health impact and the extent of intervention.

Research Question 3. Quarantine level sensitivity analysis

Sensitivity of Total Deaths to Quarantine Level



The sensitivity analysis indicates that a quarantine level of approximately 39% is the most effective in minimizing total deaths. This suggests that quarantine, on its own, can be sufficient to prevent Disease X from progressing into a widespread epidemic and significantly reduce mortality. However, implementing and enforcing quarantine measures at this scale can be logistically and socially challenging.

Therefore, we recommend adopting a quarantine level of around 38% in combination with additional preventive strategies—such as mask-wearing, vaccination, and social distancing—to further reduce population exposure and enhance overall outbreak control.

Model 2 - lost immunity after 180 days

This is the compartmental model scheme of Disease X, where the immunity is lost after 180 days.

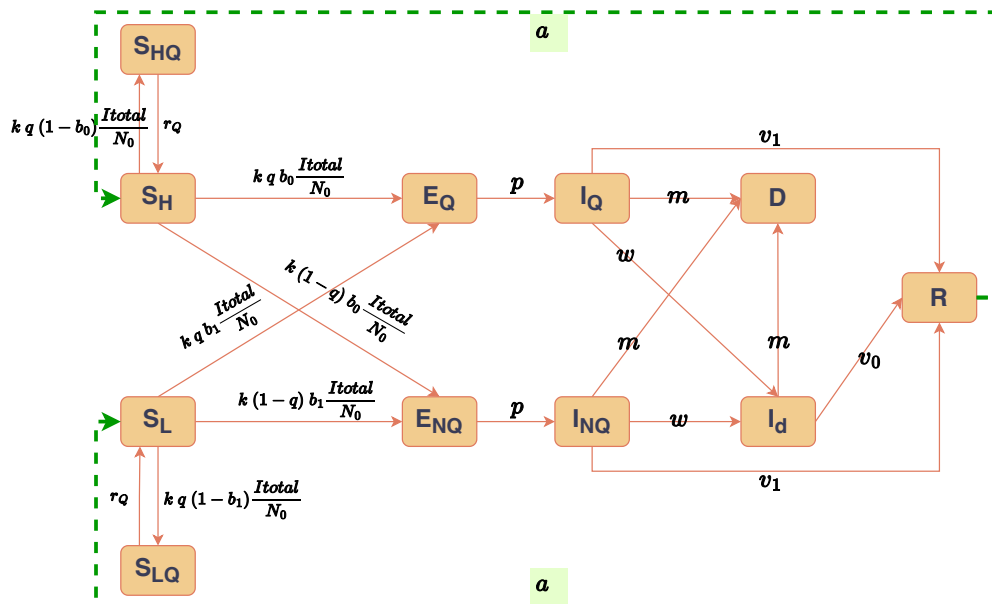
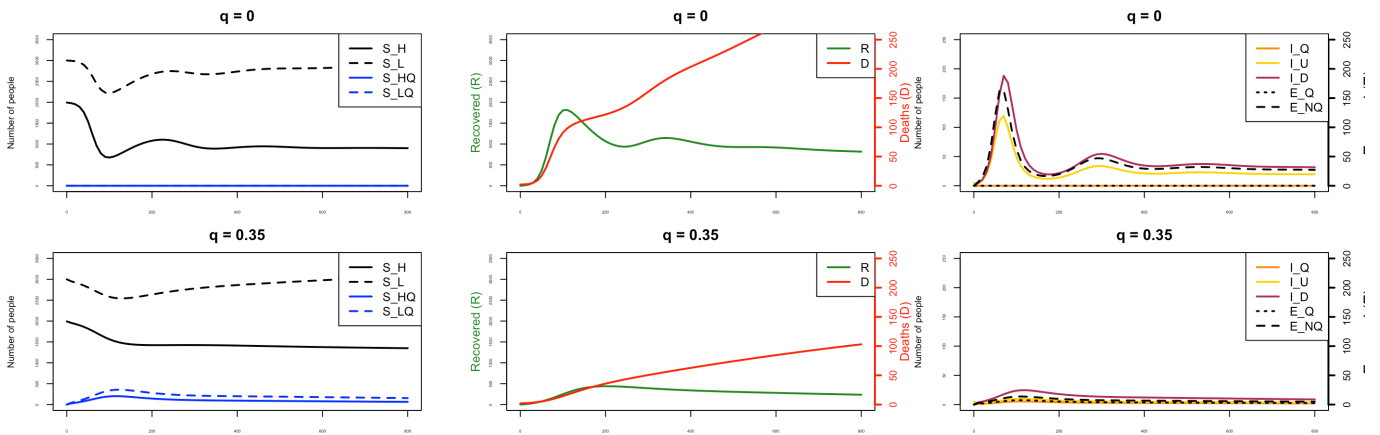


Fig 2. Model Scheme of Disease X with lost immunity after 180 days

$$\text{Total infectious: } I_{\text{total}} = 0.10 \cdot E_{\text{NQ}} + I_{\text{U}} + 0.38 \cdot I_{\text{D}}$$

1. $\frac{dS_{\text{HQ}}}{dt} = k(1 - b_0)q \cdot \frac{S_{\text{H}}I_{\text{total}}}{N_0} - r_Q S_{\text{HQ}}$
2. $\frac{dS_{\text{H}}}{dt} = \alpha R + r_Q S_{\text{HQ}} - kq(1 - b_0) \cdot \frac{S_{\text{H}}I_{\text{total}}}{N_0} - k(1 - q)b_0 \cdot \frac{S_{\text{H}}I_{\text{total}}}{N_0} - kqb_0 \cdot \frac{S_{\text{H}}I_{\text{total}}}{N_0}$
3. $\frac{dS_{\text{LQ}}}{dt} = k(1 - b_1)q \cdot \frac{S_{\text{L}}I_{\text{total}}}{N_0} - r_Q S_{\text{LQ}}$
4. $\frac{dS_{\text{L}}}{dt} = \alpha R + r_Q S_{\text{LQ}} - kq(1 - b_1) \cdot \frac{S_{\text{L}}I_{\text{total}}}{N_0} - k(1 - q)b_1 \cdot \frac{S_{\text{L}}I_{\text{total}}}{N_0} - kqb_1 \cdot \frac{S_{\text{L}}I_{\text{total}}}{N_0}$
5. $\frac{dE_{\text{Q}}}{dt} = kqb_0 \cdot \frac{S_{\text{H}}I_{\text{total}}}{N_0} + kqb_1 \cdot \frac{S_{\text{L}}I_{\text{total}}}{N_0} - pE_{\text{Q}}$
6. $\frac{dE_{\text{NQ}}}{dt} = k(1 - q)b_0 \cdot \frac{S_{\text{H}}I_{\text{total}}}{N_0} + k(1 - q)b_1 \cdot \frac{S_{\text{L}}I_{\text{total}}}{N_0} - pE_{\text{NQ}}$
7. $\frac{dI_{\text{Q}}}{dt} = pE_{\text{Q}} - I_{\text{Q}}(m + w_1 + v_0)$
8. $\frac{dI_{\text{U}}}{dt} = pE_{\text{NQ}} - I_{\text{U}}(m + w_2 + v_0)$
9. $\frac{dI_{\text{D}}}{dt} = w_1 I_{\text{Q}} + w_2 I_{\text{U}} - mI_{\text{D}} - v_1 I_{\text{D}}$
10. $\frac{dD}{dt} = m(I_{\text{Q}} + I_{\text{U}} + I_{\text{D}})$
11. $\frac{dR}{dt} = v_0(I_{\text{Q}} + I_{\text{U}}) + v_1 I_{\text{D}} - 2\alpha R$



When immunity wanes after 180 days and the population becomes susceptible again, the epidemic begins earlier and, without any quarantine measures, it does not end naturally as it does in model 1. The death toll continues to rise, and equilibrium—the end of the epidemic—is not reached even after 800 days.

With a quarantine level of 35%, equilibrium is eventually achieved, and the overall size of the epidemic is smaller. However, the death toll still rises steadily in a similar pattern.

Introducing quarantine in this scenario has a significant impact, greatly reducing the damage caused by the epidemic.

Conclusions

Introducing quarantine has a significant impact in preventing the spread of Disease X, as demonstrated by all models across different scenarios. Based on sensitivity analysis, we recommend adopting a quarantine level of approximately 38%, combined with additional preventive measures—such as mask-wearing, vaccination, and social distancing—to further reduce population exposure and enhance overall outbreak control.